MIL-F-24646/16(SH)
30 November 1984
SUPERSEDING
(See table I)

MILITARY SPECIFICATION SHEET

FUSE BOXES, FEEDER JUNCTION, SUBMERSIBLE (15 FEET), 250 VOLTS, ALTERNATING CURRENT, SINGLE PHASE BUSES, 0-30, 31-60, 61-100 AND 101-200 AMPERE FUSES, SYMBOLS 687.1, 688.1, 689.1 AND 690.1

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fuse boxes described herein shall consist of this document and the latest issue of MIL-F-24646(SH).

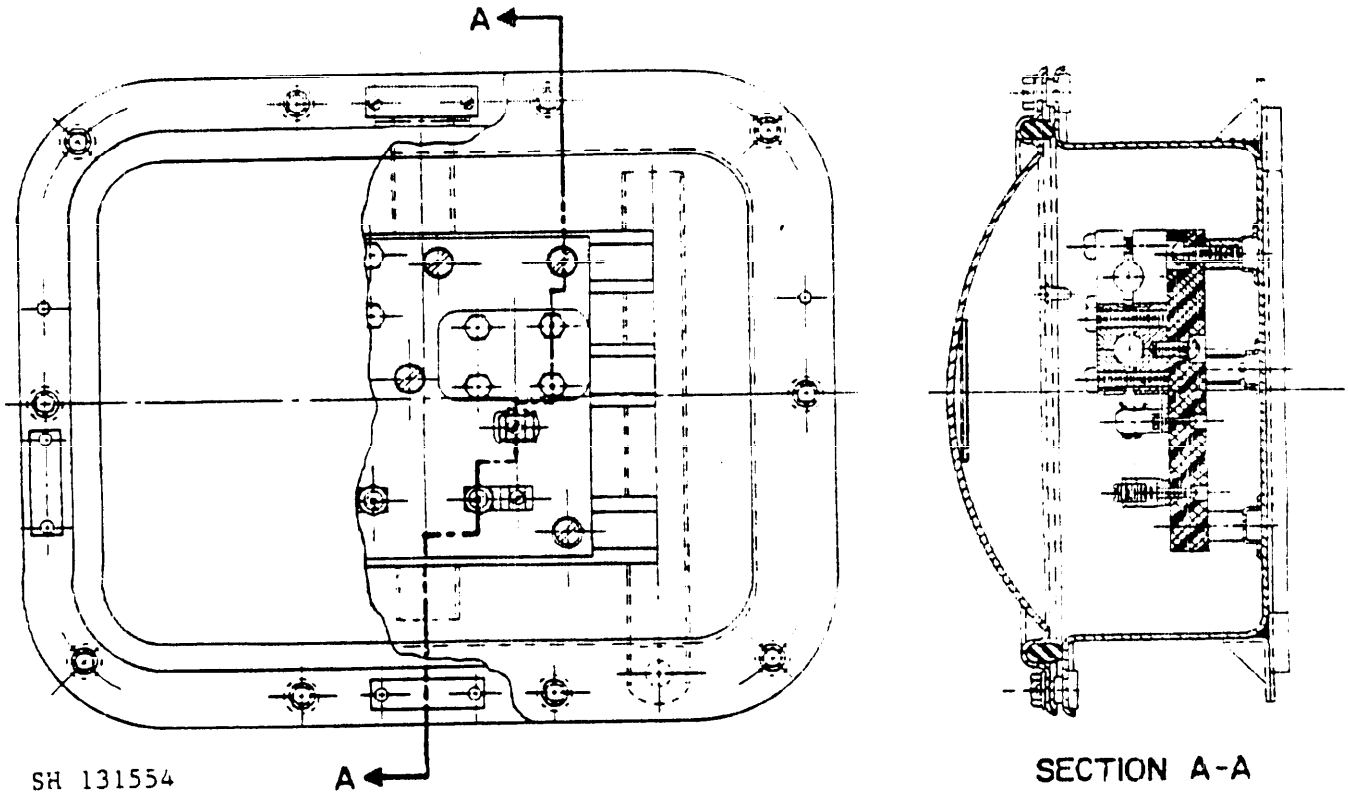


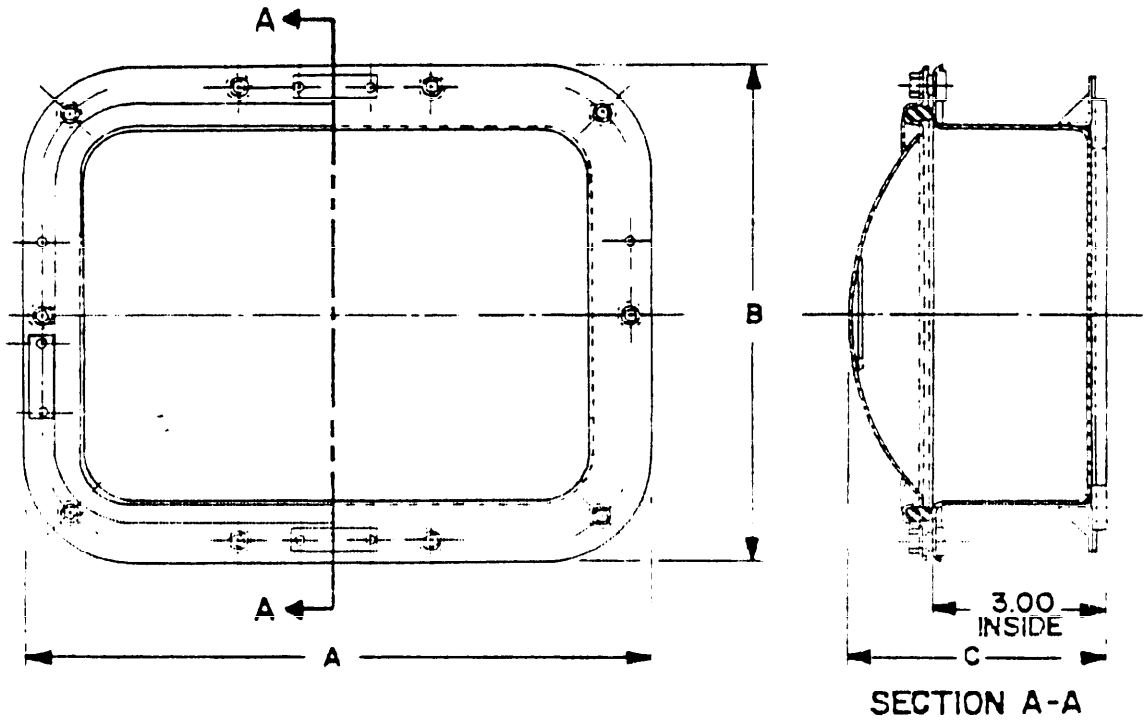
FIGURE 1. Fuse box assembly, typical. (01) (02) (03) (04)
(Symbol 687.1 shown)

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified herein, tolerances shall be as follows:
   Two place decimals - plus or minus 0.02.
   Three place decimals - plus or minus 0.005.

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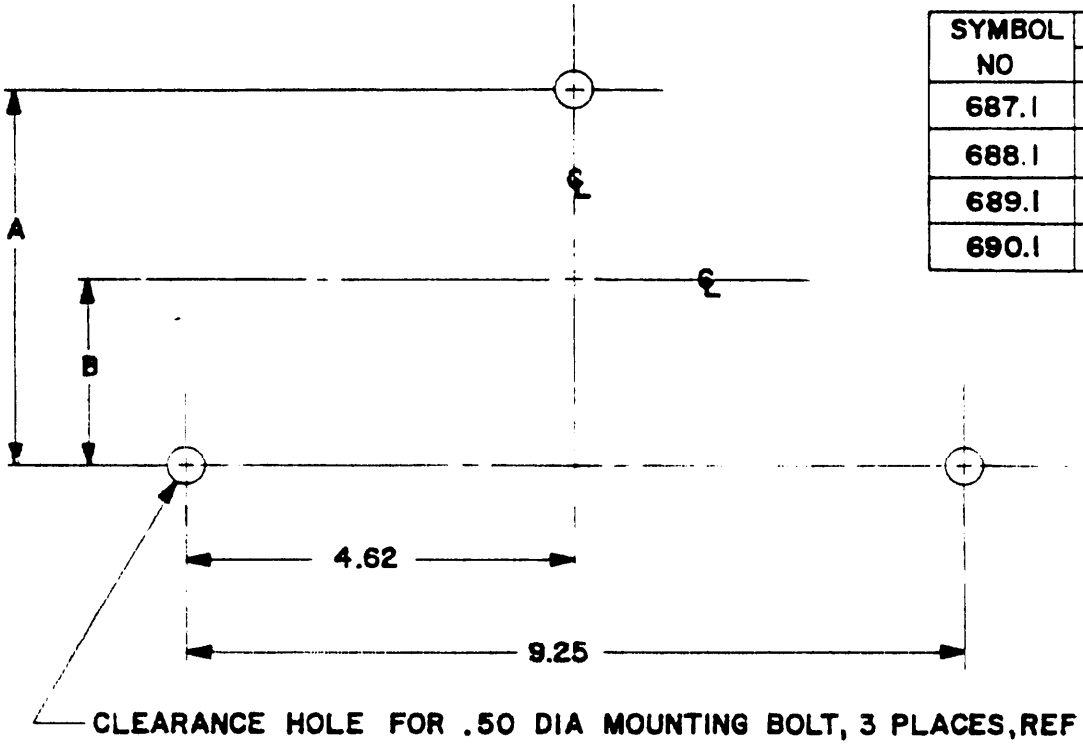


SYMBOL NO	DIMENSIONS (REF)		
	A	B	C
687.1	16.38	13.38	7.44
688.1	16.38	13.38	7.44
689.1	16.38	15.38	7.44
690.1	16.38	15.38	7.44

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FIGURE 2. Enclosure, overall dimensions. (05) (06)
(Symbols 687.1, 688.1, 689.1 and 690.1)

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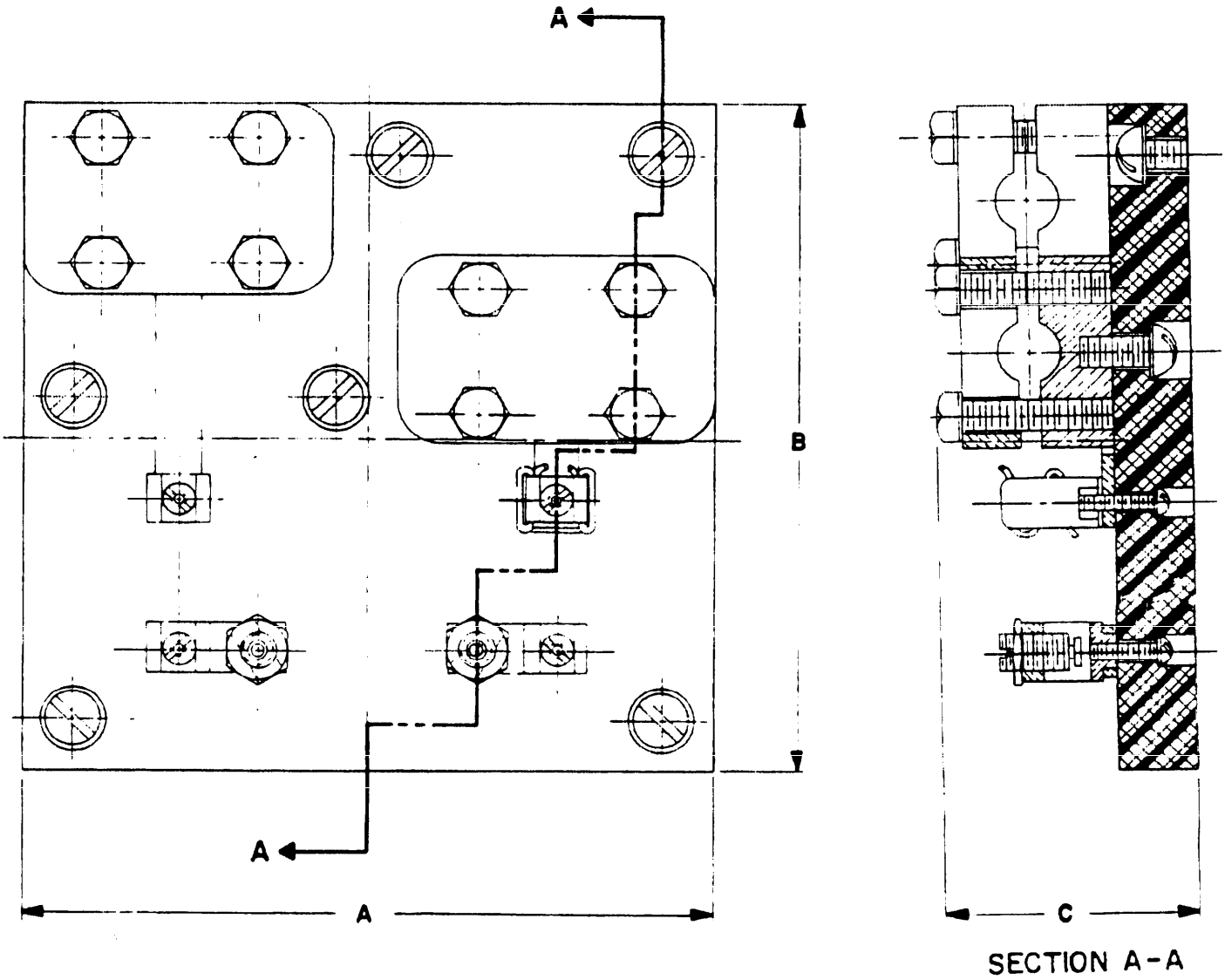
SYMBOL NO	DIMENSIONS A	B
687.1	11.44	5.72
688.1	11.44	5.72
689.1	13.44	6.72
690.1	13.44	6.72

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FIGURE 3. Mounting dimensions, ref.

(Symbols 687.1, 688.1, 689.1 and 690.1)

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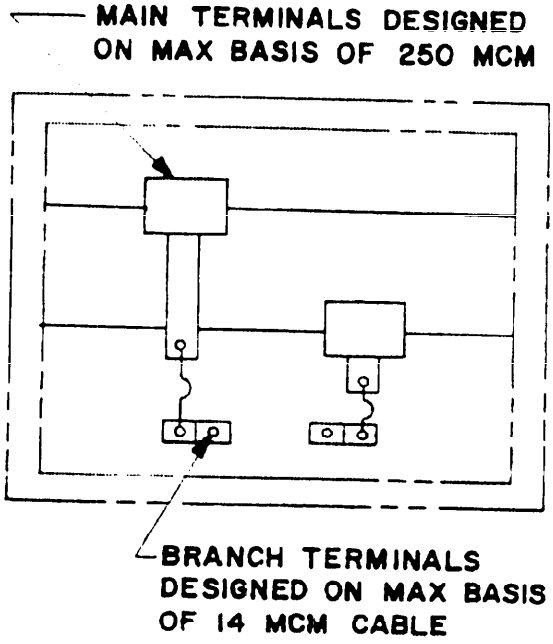
SYMBOL NO	DIMENSIONS (MAX)		
	A	B	C
687.1	6.62	6.62	2.42
688.1	6.62	6.62	2.42
689.1	7.50	6.50	4.25
690.1	7.50	7.00	4.56

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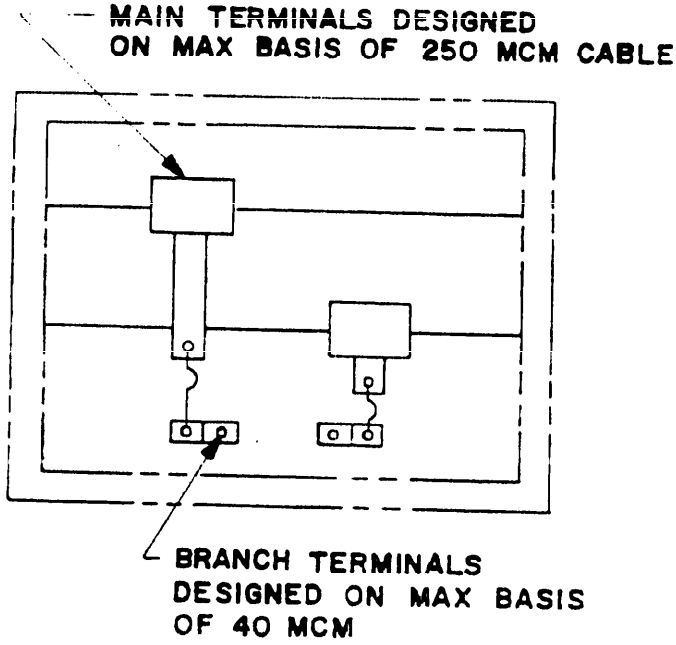
FIGURE 4. Fuse block assembly, typical. (07) (08) (09) (10)

(Symbols 687.1, 688.1, 689.1 and 690.1)

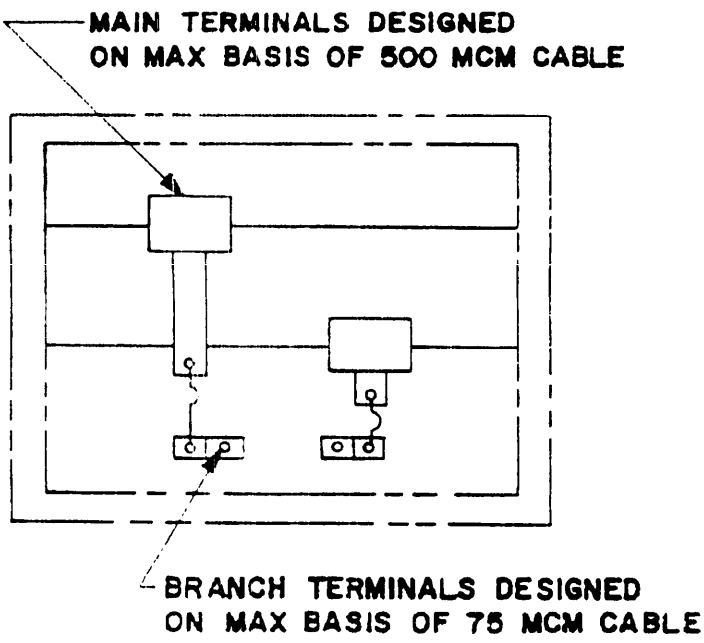
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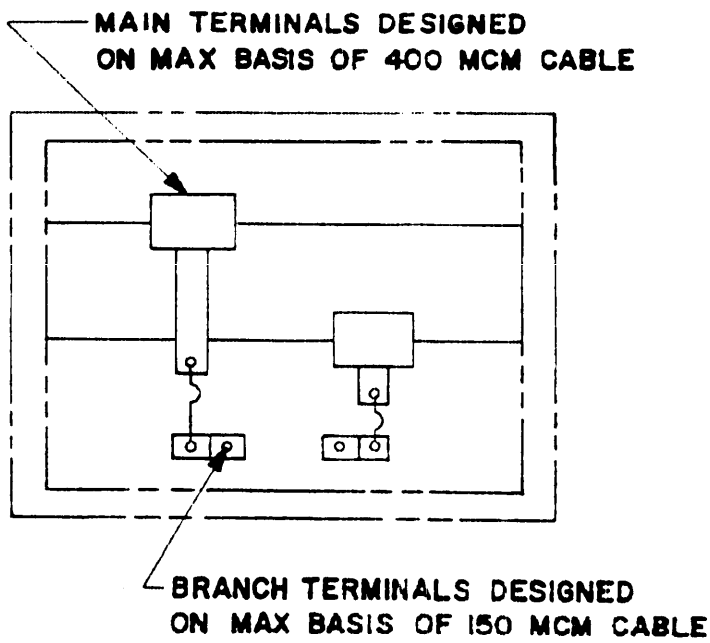
A. One single-phase branch
(Symbol 687.1)



B. One single-phase branch
(Symbol 688.1)



C. One single-phase branch
(Symbol 689.1)



D. One single-phase branch
(Symbol 690.1)

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FIGURE 5. Wiring diagrams. (11) (12) (13) (14)

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REQUIREMENTS:

Symbol numbers: See table I.

Design:
Electrical rating: 250 volts alternating current, 0-30, 31-60, 61-100, and 101-200 ampere.
Dimensions and configuration: See figures 1, 2, 3 and 4.
Wiring diagrams: See figure 5.
Mounting dimensions: See figure 3.
Degree of enclosure: Submersible (15 feet).
Designation and marking: See MIL-F-24646(SH).
Parts: The parts listed in table II and specified in MIL-F-24646(SH) shall be used in the construction of the fuse boxes specified herein.

General information:
Military-part numbers and general information are as specified in table I.

TABLE I. Military part numbers and general information.

Military part no.	Symbol no.	Weight in pounds	No. of circuits	Feeder MCM	Branch MCM	Fuse size	Bus bar size	Supersedes	
								Drawing no.	Type
M24646/16-001	687.1		1	250	14	0-30A	0.50 x 0.12	9000-S-74184	F-52A
M24646/16-002	688.1		1	250	40	31-60A	.75 x .25	9000-S-74186	F-33A
M24646/16-003	689.1	36.0	1	500	75	61-100A	.88 x .25	9000-S-74187	F-35A
M24646/16-004	690.1		1	400	150	101-200A	1.00 x .38	9000-S-74188	F-27A

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TABLE II. Parts list.

Item no. 1/	Description	QTY M /-				Material/spec	Remarks
		001	002	003	004		
01	Fuse box assembly (sym 687.1)	-				MIL-F-24646/16	See figure 1
02	Fuse box assembly (sym 688.1)		-			MIL-F-24646/16	See figure 1
03	Fuse box assembly (sym 689.1)			-		MIL-F-24646/16	See figure 1
04	Fuse box assembly (sym 690.1)				-	MIL-F-24646/16	See figure 1
2/05	Enclosure (10 x 13 x 4)	1	1			M24142/6-004	See figure 2 Type A
06	Enclosure (12 x 13 x 5)			1		M24142/6-008	See figure 2 Type A
07	Fuse block assembly	1				MIL-F-24646/16	See figure 4
08	Fuse block assembly		1			MIL-F-24646/16	See figure 4
09	Fuse block assembly			1		MIL-F-24646/16	See figure 4
10	Fuse block assembly				1	MIL-F-24646/16	See figure 4
11	Wiring diagram	1				MIL-E-2036	See figure 5
12	Wiring diagram		1			MIL-E-2036	See figure 5
13	Wiring diagram			1		MIL-E-2036	See figure 5
14	Wiring diagram				1	MIL-E-2036	See figure 5
15	Fuse clip (0-30A)	4	-	-	-	M21346/1-06	----
16	Fuse clip (31-60A)	-	4	-	-	M21346/1-08	----
17	Fuse clip with retainer (61-100A)			4			4319-L16
18	Fuse clip with retainer (101-200A)				4		4360-L16
19	Main terminal assembly	2	2	2	2	NAVSEA/SHIPS 9000-S-73275	
20	Identification plate	1	1	1	1	Brass, QQ-B-613	----
21	Information plate	1	1	1	1	Brass, QQ-B-613	----

1/ Item numbers are the same as numbers shown in circles on figures.
2/ Enclosure is same as MIL-E-24142/6(SH), except for inside depth which is 4 inches instead of 5 shown.

Preparing activity:
Navy - SH
(Project 5920-N163-16)

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